

INTERNSHIP PROJECT REPORT

**DATA CLEANING AND SALES DATA
ANALYSIS USING EXCEL, PYTHON, SQL
AND POWER BI**

SUBMITTED BY : PRATYUSHA SARKAR

**INTERNSHIP NAME : DATA ANALYST
INTERNSHIP**

ORGANIZATION : CODEORBIT TECH

INTERNSHIP DURATION : 1.5 MONTHS

ACADEMIC YEAR : 2026

OBJECTIVE :

The goal of this project is so clean and analyze sales data using Excel and Python and extract useful business insights from the data. The project also aims to present the results through charts and dashboard, and SQL queries for business insights in a simple and understandable way.

DATASET :

I download the sales data set from Online Retail Dataset of kaggle's website. This dataset is a CSV file.

DATA CLEANING :

In this step data cleaning I used to excel and handled missing values, and corrected the format which all steps are documented and attached this.

PYTHON ANALYSIS :

This step I used in Pandas and calculate Total sales, Top 10 and 5 countries according total sales from largest to smallest, Top 10 customers by sales and Top 10 products by sales and create bar chart and pie. All steps are documented and attached this.

DASHBOARD :

I create dashboard in excel using Pivot table. Here the dashboard

1. All details of Pivot Table are sorted in largest to smallest.
2. All details in Dashboard.
3. Top Quantity of Product in Dashboard.

1. Create Pivot Table

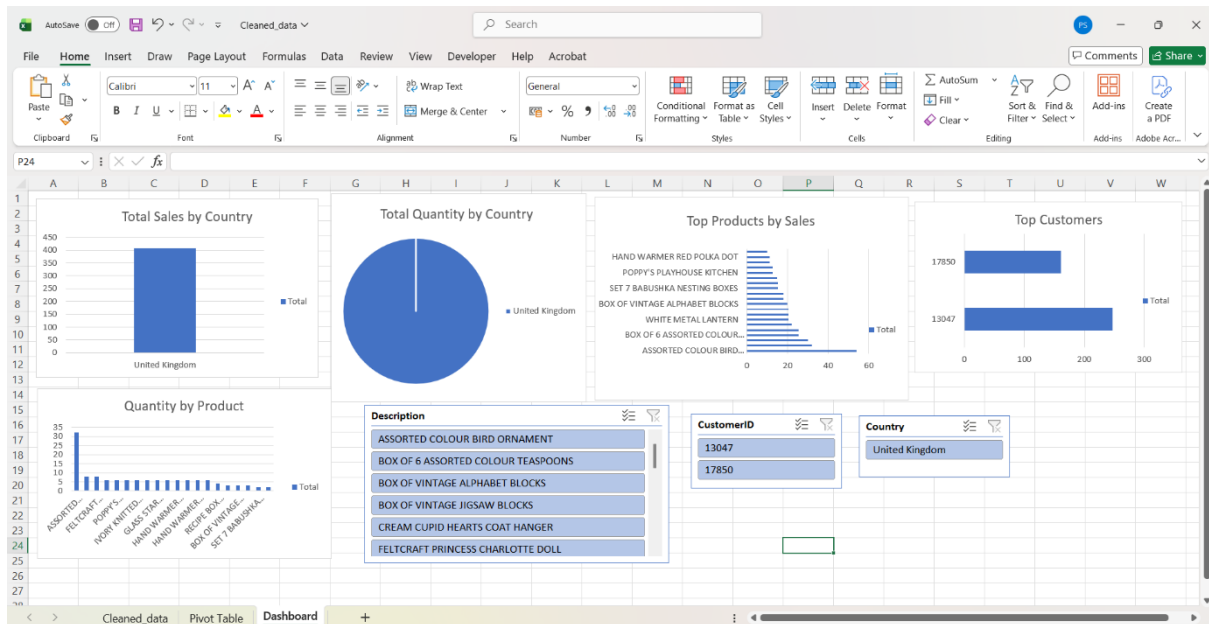
The screenshot shows an Excel PivotTable with the following data:

Row Labels	Sum of Total Sales	Row Labels	Sum of Quantity
United Kingdom	408.25	United Kingdom	131
Grand Total	408.25	Grand Total	131

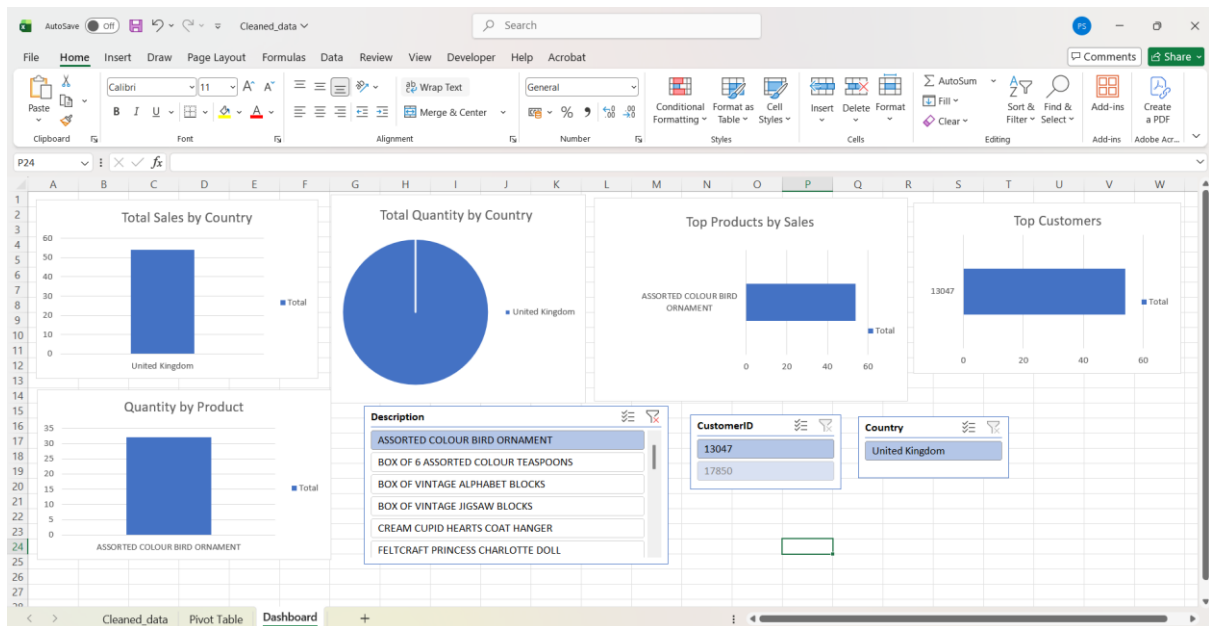
The PivotTable is based on the following data (from the adjacent table):

Row Labels	Sum of Total Sales	Row Labels	Sum of Quantity
ASSORTED COLOUR BIRD ORNAMENT	54.08	ASSORTED COLOUR BIRD ORNAMENT	32
RECIPE BOX WITH METAL HEART	31.8	RECIPE BOX WITH METAL HEART	8
FELTCRAFT PRINCESS CHARLOTTE DOLL	30	FELTCRAFT PRINCESS CHARLOTTE DOLL	8
BOX OF 6 ASSORTED COLOUR TEASPOONS	25.5	RED WOOLLY HOTTIE WHITE HEART	6
GLASS STAR FROSTED T-LIGHT HOLDER	25.5	POPPY'S PLAYHOUSE BEDROOM	6
CREAM CUPID HEARTS COAT HANGER	22	BOX OF 6 ASSORTED COLOUR TEASPOONS	6
WHITE METAL LANTERN	20.34	IVORY KNITTED MUG COSY	6
RED WOOLLY HOTTIE WHITE HEART	20.34	KNITTED UNION FLAG HOT WATER BOTTLE	6
KNITTED UNION FLAG HOT WATER BOTTLE	20.34	GLASS STAR FROSTED T-LIGHT HOLDER	6
BOX OF VINTAGE ALPHABET BLOCKS	19.9	POPPY'S PLAYHOUSE KITCHEN	6
LOVE BUILDING BLOCK WORD	17.85	HAND WARMER RED POLKA DOT	6
HOME BUILDING BLOCK WORD	17.85	WHITE HANGING HEART T-LIGHT HOLDER	6
SET 7 BABUSHKA NESTING BOXES	15.3	HAND WARMER UNION JACK	6
WHITE HANGING HEART T-LIGHT HOLDER	15.3	WHITE METAL LANTERN	6
BOX OF VINTAGE JIGSAW BLOCKS	14.85	RECIPE BOX WITH METAL HEART	4
POPPY'S PLAYHOUSE KITCHEN	12.6	LOVE BUILDING BLOCK WORD	3
POPPY'S PLAYHOUSE BEDROOM	12.6	BOX OF VINTAGE JIGSAW BLOCKS	3
HAND WARMER UNION JACK	11.1	HOME BUILDING BLOCK WORD	3
HAND WARMER RED POLKA DOT	11.1	SET 7 BABUSHKA NESTING BOXES	2
IVORY KNITTED MUG COSY	9.9	BOX OF VINTAGE ALPHABET BLOCKS	2
Grand Total	408.25	Grand Total	131

2. All details in Dashboard



3. Top Quantity of Product in Dashboard



SQL QUERIES :

In this step I used Sql Server Management Studio and find all the business insights. All steps are documented and attached this. Here the important Queries.

1. Find Sales by Country

The screenshot shows the SQL Server Management Studio interface with a list of queries and their results. The queries are as follows:

```

select Description,UnitPrice from Cleaned_data where UnitPrice >5;
select CustomerID ,Country from Cleaned_data where Country='United Kingdom';
select Description,Sum(Quantity*UnitPrice) as TotalSales from Cleaned_data group by Description;
select Top 10 Description,Sum(Quantity) as TotalQuantity from Cleaned_data group by Description order by TotalQuantity Desc;
select count(Distinct InvoiceNo) as TotalOrders from Cleaned_data;
select Sum(Quantity*UnitPrice) as TotalRevenue from Cleaned_data;
select Avg(UnitPrice) as AveragePrice from Cleaned_data;
select Top 1 CustomerID, Sum(Quantity) as TotalQuantity from Cleaned_data group by CustomerID order by TotalQuantity Desc;
select Country,Sum(Quantity*UnitPrice) as TotalSales from Cleaned_data group by Country order by TotalSales Desc;
select Top 1 Description, UnitPrice from Cleaned_data order by UnitPrice Desc;
select Description,Quantity from Cleaned_data where quantity > 10;
select Country ,Count(Distinct CustomerID) from Cleaned_data group by Country;
select InvoiceNo,Count(*) as Total from Cleaned_data group by InvoiceNo Having Count(*) > 1;
select Top 5 CustomerID,Sum(Quantity*UnitPrice) as Revenue from Cleaned_data group by CustomerID order by Revenue Desc;

```

The results pane shows the output of the query: `select Country,Sum(Quantity*UnitPrice) as TotalSales from Cleaned_data group by Country order by TotalSales Desc;`

Country	TotalSales
1 United Kingdom	408.25

Query executed successfully.

2. Top Quantity of Product

The screenshot shows the Microsoft SQL Server Management Studio interface. The query editor contains a series of SQL statements. The last statement is highlighted, which is a query to find the top product by quantity. The results pane shows a single row with the product description and its quantity.

```
select Description,UnitPrice from Cleaned_data where UnitPrice >5;
select CustomerID ,Country from Cleaned_data where Country='United Kingdom';
select Description,Sum(Quantity*UnitPrice) as TotalSales from Cleaned_data group by Description;
select Top 10 Description,Sum(Quantity) as TotalQuantity from Cleaned_data group by Description order by TotalQuantity Desc;
select count(Distinct InvoiceNo) as TotalOrders from Cleaned_data;
select Sum(Quantity*UnitPrice) as TotalRevenue from Cleaned_data;
select Avg(UnitPrice) as AveragePrice from Cleaned_data;
select Top 1 CustomerID, Sum(Quantity) as TotalQuantity from Cleaned_data group by CustomerID order by TotalQuantity Desc;
select Country,Sum(Quantity*UnitPrice) as TotalSales from Cleaned_data group by Country order by TotalSales Desc;
select Top 1 Description, UnitPrice from Cleaned_data order by UnitPrice Desc;
select Description,Quantity from Cleaned_data where quantity > 10;
select Country ,Count(Distinct CustomerID) from Cleaned_data group by Country;
select InvoiceNo,Count(*) as Total from Cleaned_data group by InvoiceNo Having Count(*) > 1;
select Top 5 CustomerID,Sum(Quantity*UnitPrice) as Revenue from Cleaned_data group by CustomerID order by Revenue Desc;
```

Description	Quantity
ASSORTED COLOUR BIRD ORNAMENT	32

Query executed successfully. LAPTOP-TCEV345R\SQLEXPRESS ... LAPTOP-TCEV345R\User (90) Sales 00:00:00 1 rows

3. The highest quantity of purchased customer.

The screenshot shows the Microsoft SQL Server Management Studio interface. The query editor contains the same series of SQL statements as in the previous screenshot. The last statement is highlighted, which is a query to find the customer with the highest total quantity purchased. The results pane shows a single row with the customer ID and their total quantity.

```
select Description,UnitPrice from Cleaned_data where UnitPrice >5;
select CustomerID ,Country from Cleaned_data where Country='United Kingdom';
select Description,Sum(Quantity*UnitPrice) as TotalSales from Cleaned_data group by Description;
select Top 10 Description,Sum(Quantity) as TotalQuantity from Cleaned_data group by Description order by TotalQuantity Desc;
select count(Distinct InvoiceNo) as TotalOrders from Cleaned_data;
select Sum(Quantity*UnitPrice) as TotalRevenue from Cleaned_data;
select Avg(UnitPrice) as AveragePrice from Cleaned_data;
select Top 1 CustomerID, Sum(Quantity) as TotalQuantity from Cleaned_data group by CustomerID order by TotalQuantity Desc;
select Country,Sum(Quantity*UnitPrice) as TotalSales from Cleaned_data group by Country order by TotalSales Desc;
select Top 1 Description, UnitPrice from Cleaned_data order by UnitPrice Desc;
select Description,Quantity from Cleaned_data where quantity > 10;
select Country ,Count(Distinct CustomerID) from Cleaned_data group by Country;
select InvoiceNo,Count(*) as Total from Cleaned_data group by InvoiceNo Having Count(*) > 1;
select Top 5 CustomerID,Sum(Quantity*UnitPrice) as Revenue from Cleaned_data group by CustomerID order by Revenue Desc;
```

CustomerID	TotalQuantity
13047	79

Query executed successfully. LAPTOP-TCEV345R\SQLEXPRESS ... LAPTOP-TCEV345R\User (90) Sales 00:00:00 1 rows

4. Find the most expensive product.

The screenshot shows the Microsoft SQL Server Management Studio interface. The query editor contains a SQL query that filters for products with a unit price greater than 5 and returns the top 1 result ordered by unit price in descending order. The query is as follows:

```
select Description,UnitPrice from Cleaned_data where UnitPrice >5;
select CustomerID ,Country from Cleaned_data where Country='United Kingdom';
select Description,Sum(Quantity*UnitPrice) as TotalSales from Cleaned_data group by Description;
select Top 10 Description,Sum(Quantity) as TotalQuantity from Cleaned_data group by Description order by TotalQuantity Desc;
select count(Distinct InvoiceNo) as TotalOrders from Cleaned_data;
select Sum(Quantity*UnitPrice) as TotalRevenue from Cleaned_data;
select Avg(UnitPrice) as AveragePrice from Cleaned_data;
select Top 1 CustomerID, Sum(Quantity) as TotalQuantity from Cleaned_data group by CustomerID order by TotalQuantity Desc;
select Country,Sum(Quantity*UnitPrice) as TotalSales from Cleaned_data group by Country order by TotalSales Desc;
select Top 1 Description, UnitPrice from Cleaned_data order by UnitPrice Desc;
select Description,Quantity from Cleaned_data where quantity > 10;
select Country ,Count(Distinct CustomerID) from Cleaned_data group by Country;
select InvoiceNo,Count(*) as Total from Cleaned_data group by InvoiceNo Having Count(*) > 1;
select Top 5 CustomerID,Sum(Quantity*UnitPrice) as Revenue from Cleaned_data group by CustomerID order by Revenue Desc;
```

The Results pane shows the following data:

Description	UnitPrice
BOX OF VINTAGE ALPHABET BLOCKS	9.95

The status bar at the bottom indicates "Query executed successfully." and "1 rows".

5. Find top 5 customers by revenue

The screenshot shows the Microsoft SQL Server Management Studio interface. The query editor contains a SQL query that filters for products with a unit price greater than 5 and returns the top 5 results ordered by revenue in descending order. The query is as follows:

```
select Description,UnitPrice from Cleaned_data where UnitPrice >5;
select CustomerID ,Country from Cleaned_data where Country='United Kingdom';
select Description,Sum(Quantity*UnitPrice) as TotalSales from Cleaned_data group by Description;
select Top 10 Description,Sum(Quantity) as TotalQuantity from Cleaned_data group by Description order by TotalQuantity Desc;
select count(Distinct InvoiceNo) as TotalOrders from Cleaned_data;
select Sum(Quantity*UnitPrice) as TotalRevenue from Cleaned_data;
select Avg(UnitPrice) as AveragePrice from Cleaned_data;
select Top 1 CustomerID, Sum(Quantity) as TotalQuantity from Cleaned_data group by CustomerID order by TotalQuantity Desc;
select Country,Sum(Quantity*UnitPrice) as TotalSales from Cleaned_data group by Country order by TotalSales Desc;
select Top 1 Description, UnitPrice from Cleaned_data order by UnitPrice Desc;
select Description,Quantity from Cleaned_data where quantity > 10;
select Country ,Count(Distinct CustomerID) from Cleaned_data group by Country;
select InvoiceNo,Count(*) as Total from Cleaned_data group by InvoiceNo Having Count(*) > 1;
select Top 5 CustomerID,Sum(Quantity*UnitPrice) as Revenue from Cleaned_data group by CustomerID order by Revenue Desc;
```

The Results pane shows the following data:

CustomerID	Revenue
13047	246.93
17850	161.32

The status bar at the bottom indicates "Query executed successfully." and "2 rows".

CONCLUSION :

This dataset presents various business insights; some of the important queries include which are

- 1.This dataset is include one country is United kingdom.
- 2.Top quantity of product is ASSORTED COLOUR BIRD ORNAMENT.
3. The highest quantity of purchased customer is 13047.
4. The most expensive product is BOX OF VINTAGE ALPHABET BLOCK.
- 5.Top 5 customers by revenue this are 13047,17850.

This all the details which are important to make for business growth and development.